

shiny_dashboard.R

prath

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```
# PROJECT 5: Interactive R Shiny Web Application  
# Goal: Allow dynamic filtering and plotting of plant traits.
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# packages
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library(shiny)
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## Warning: package 'shiny' was built under R version 4.5.3
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library(readxl)
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## Warning: package 'readxl' was built under R version 4.5.3
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library(ggplot2)
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## Warning: package 'ggplot2' was built under R version 4.5.3
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# 1. Read data - CHANGE THIS PATH if needed
d<- read_excel("C:/Users/prath/OneDrive/Desktop/proj1.xlsx",
               col_types = c("text", "numeric", "numeric",
                             "numeric", "numeric", "text"))

# 2. USER INTERFACE (frontend process)
ui <- fluidPage(

  titlePanel("Plant Trait Explorer"),

  # selectInput takes input from the user
  selectInput( "trait", "Choose a trait:", # "trait" stored behind the dropbox and named "Choose
a trait:" upon it
               choices = c("Height" = "L",
                           "Breadth" = "B",
                           "Shoot Length" = "SL",
                           "Root Length" = "RL")),

  # plotOutput gives output to the user
  plotOutput("myplot")
)

# 3. SERVER (Backend process)
server <- function(input, output)
{
  output$myplot <- renderPlot({

    ggplot(d, aes(x = LC, y = .data[[input$trait]], fill = LC)) +
      geom_boxplot() +
      labs(title = paste("Boxplot of", input$trait, "by Leaf Colour"),
           x = "Leaf Colour", y = input$trait) +
      theme_classic()
  })
}

# 4. Launch the app
shinyApp(ui = ui, server = server) #ui object from frontend and server objectect of backend

```

Shiny applications not supported in static R Markdown documents

